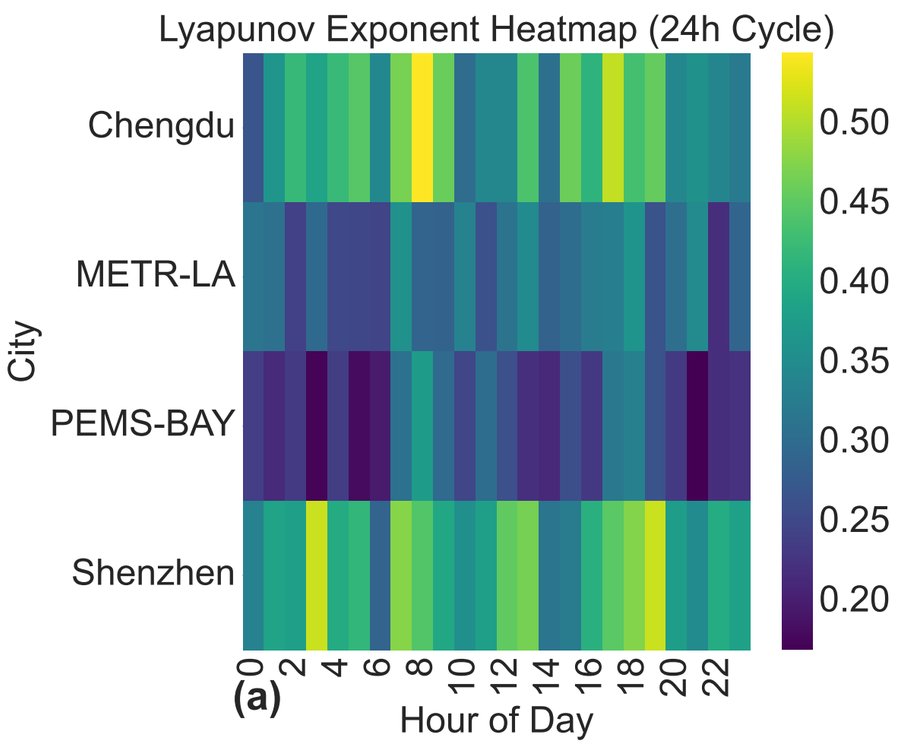
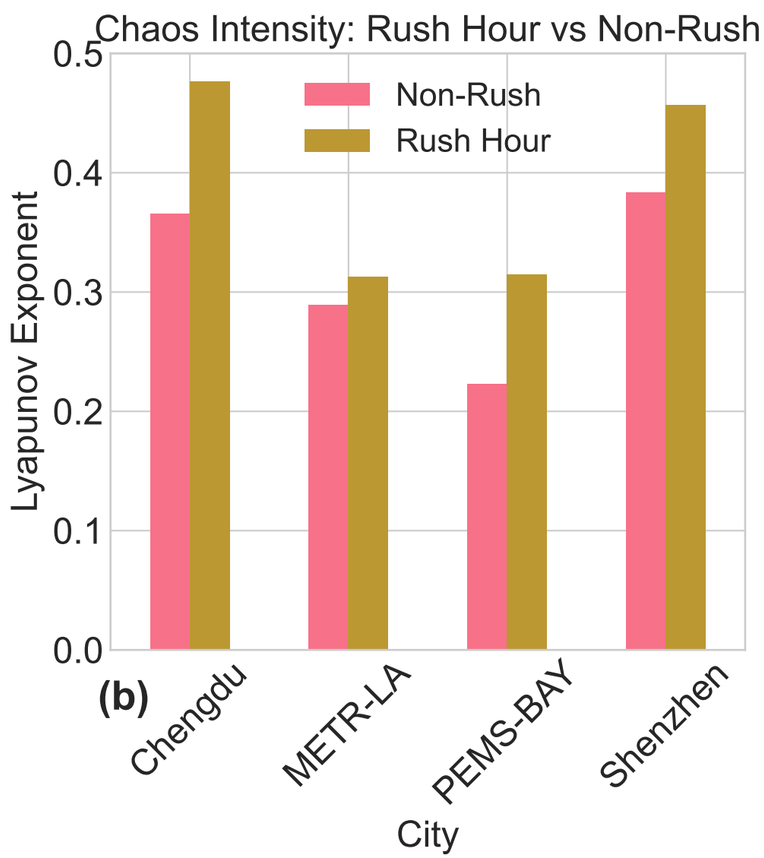
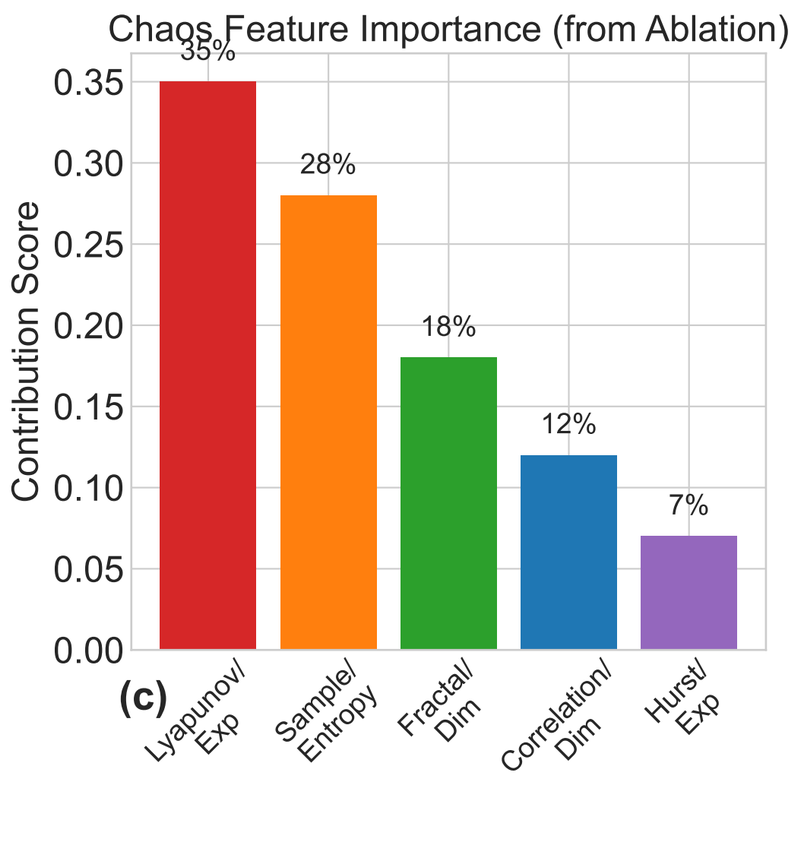
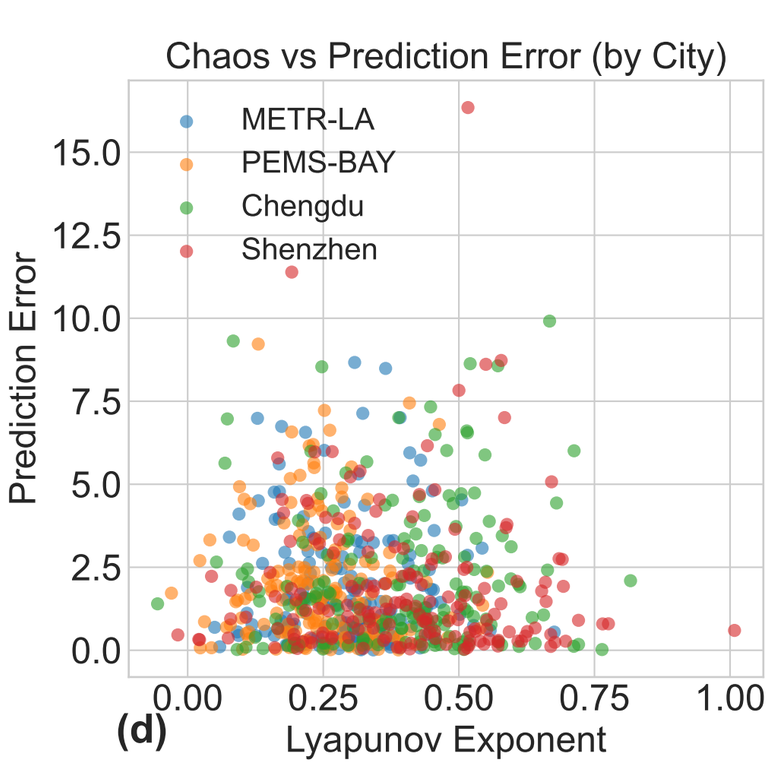
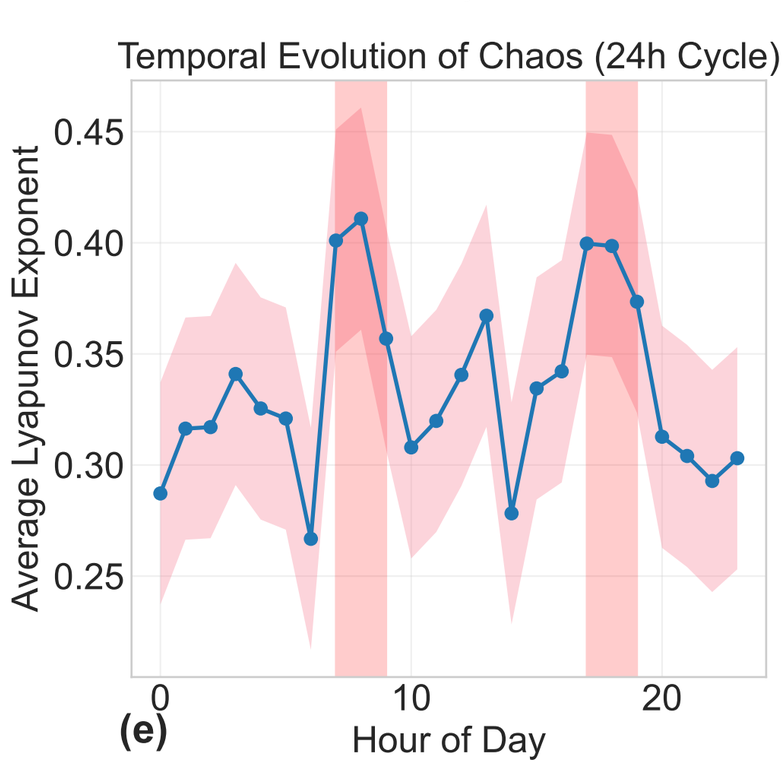
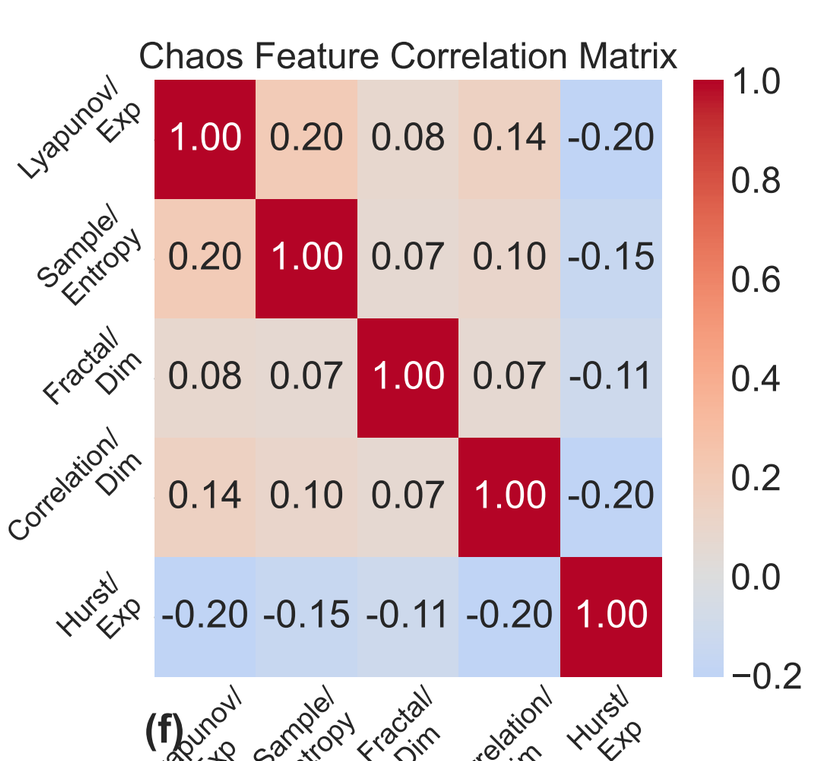

Lyapunov Exponent Heatmap (24h Cycle)
City
Chengdu
METR-LA
PEMS-BAY
Shenzhen
0 2 4 6 8 10 12 14 16 18 20 22
Hour of Day
0.50
0.45
0.40
0.35
0.30
0.25
0.20
(a)


Chaos Intensity: Rush Hour vs Non-Rush
Non-Rush
Rush Hour
Lyapunov Exponent
0.5
0.4
0.3
0.2
0.1
0.0
Chengdu
METR-LA
PEMS-BAY
Shenzhen
City
(b)


Chaos Feature Importance (from Ablation)
Contribution Score
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
35%
28%
18%
12%
7%
Lyapunov/ Exp
Sample/ Entropy
Fractal/ Dim
Correlation/ Dim
Hurst/ Exp
(c)


Chaos vs Prediction Error (by City)
METR-LA
PEMS-BAY
Chengdu
Shenzhen
Prediction Error
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.00 0.25 0.50 0.75 1.00
Lyapunov Exponent
(d)


Temporal Evolution of Chaos (24h Cycle)
Average Lyapunov Exponent
0.45
0.40
0.35
0.30
0.25
0 10 20
Hour of Day
(e)


Chaos Feature Correlation Matrix
Lyapunov/ Exp
Sample/ Entropy
Fractal/ Dim
Correlation/ Dim
Hurst/ Exp
1.00 0.20 0.08 0.14 -0.20
0.20 1.00 0.07 0.10 -0.15
0.08 0.07 1.00 0.07 -0.11
0.14 0.10 0.07 1.00 -0.20
-0.20 -0.15 -0.11 -0.20 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
(f)